

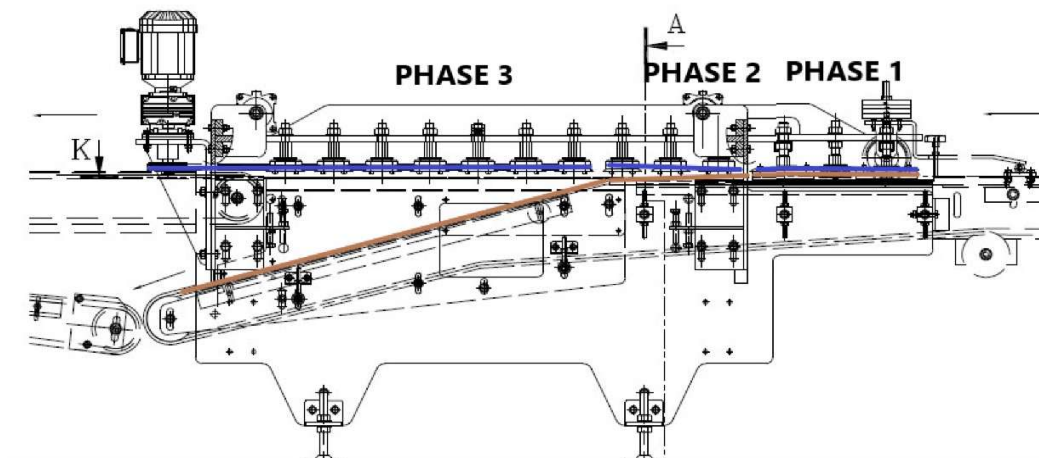
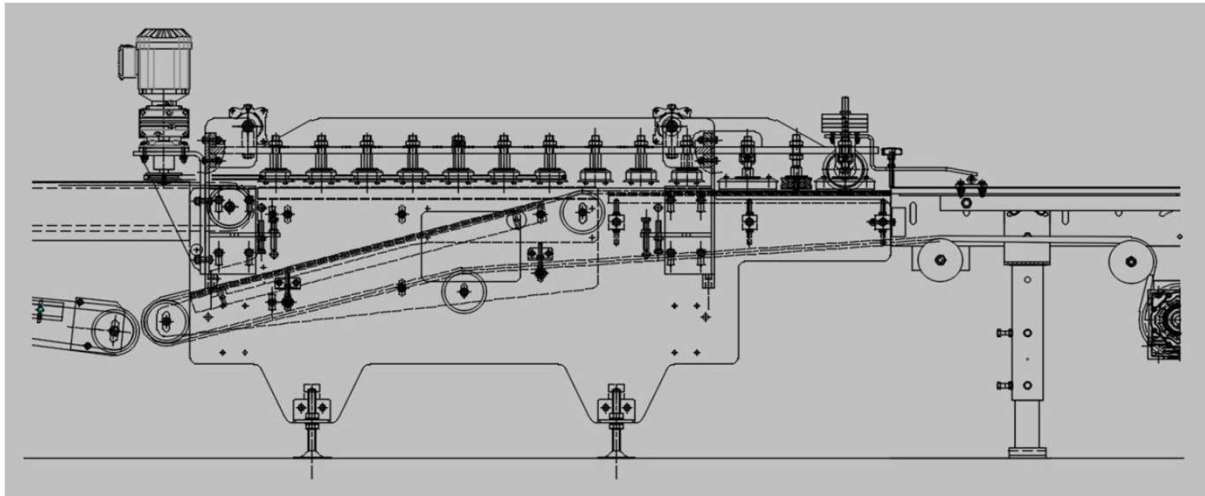
AUTOMATIC DEPALLETER -Relationship SOP

MACHINE TYPE:TM-107 SERIAL NUMBER :M.02213

FP McCANN (Cadeby)

ORDER : C22004

YEAR OF PRODUCTION 2023



SOP A.02213.00 Contents List Automatic Depalletter

SOP A.02213.01 Automatic Depalletter: Phase 1

SOP A.02213.02 Automatic Depalletter: Phase 2

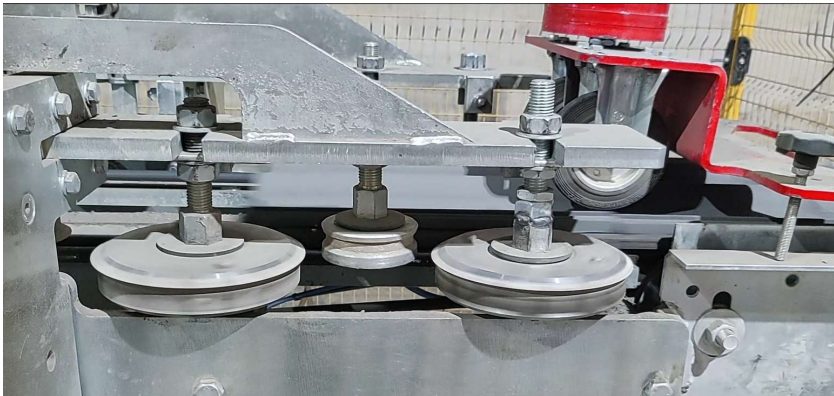
SOP A.02213.03 Automatic Depalletter: Phase 3

SOP A.02213.04 Automatic Depalletter: Positioning and Alignment

AUTOMATIC DEPALLETER

MACHINE TYPE:TM-107 SERIAL NUMBER :M.02213

Automatic Depalletter: Phase 1

**General Description and Function**

*This unit is the most critical section of the Depalletter Unit. It consists of a set of 4 blade discs (Ø200 mm) and 2 intermediate discs.

*Its function is to perform a pre-demolding operation, partially separating the tile from the mould by 2-3 mm on each side to prepare it for the final transfer phase.

Critical Points and Risk Management

*Each tile profile requires specific adjustments on the depalletter, as the mould entry height differs on both sides of the tile.

Infeed:

*The 4 towers in Phase 1, which act directly on the mould, must be repositioned.

*The towers in Phases 2 and 3 can remain in place depending on the profile.

Critical Penetration Zone:

*The overlocking area is where the tile has the strongest anchorage to the mould, making blade performance in this zone paramount.

Key Mechanical Requirement:

*Maintaining precise parallelism between the disc sets is essential for uniform separation and to prevent tile or mould damage.

Primary Wear Components:

*The 4 Ø200 mm blades are the components subject to the highest wear and must be monitored closely.

SUPERVISION LEVEL**DAILY**

5120 Disc ø200 Blade (4 units)

5124 Disc (4 units)

255561 Disc (2 units)

Visual inspection for wear and damage.

*Check the wear condition of the 4 primary blades (Ø200 mm). Refer to specific wear limits (e.g., minimum thickness, edge integrity).

WEEKLY

19500 Bearing CUS 6306 (4x2 =8 units)

17999 Bearing CUS 6205 2RS1 (2 units)

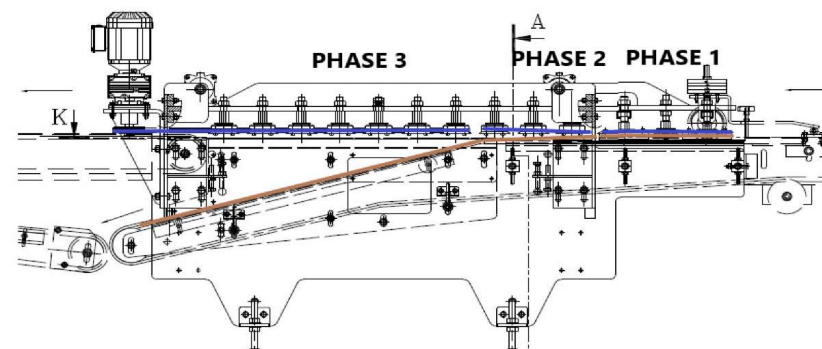
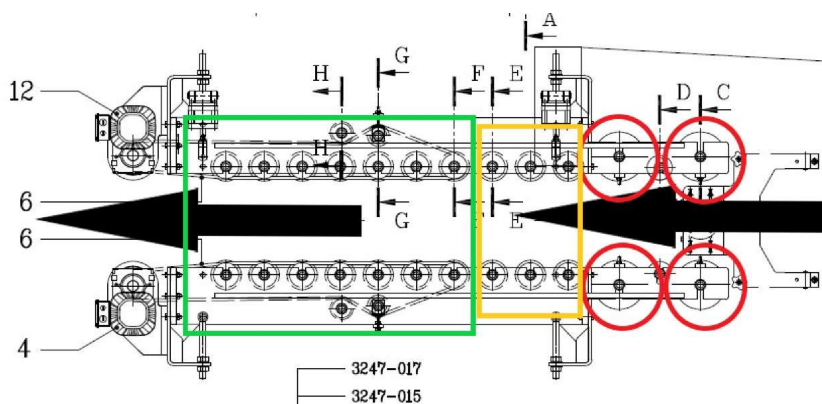
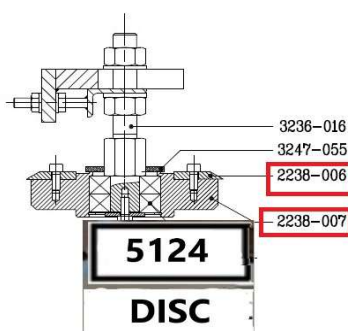
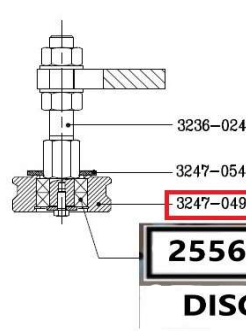
Inspection for leaks, noise, and vibration.

*Visually inspect the bearings listed under daily supervision for signs of damage or misalignment.

CLEANING**WEEKLY**

*Clean all surfaces of the structure, blades, and discs.

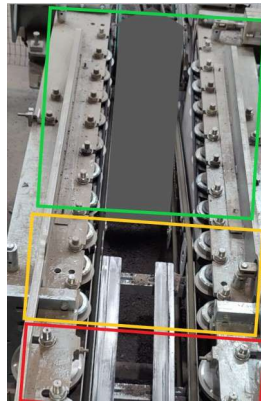
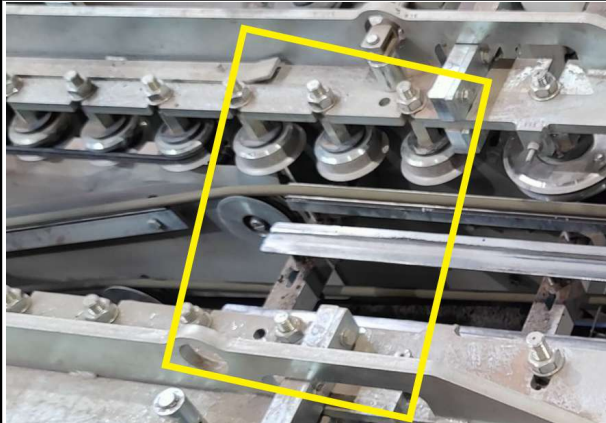
*Use a compressed air gun with low/controlled pressure and dry cloths.

**SECTION C-C****SECTION D-D**

AUTOMATIC DEPALLETTER

MACHINE TYPE: TM-107 SERIAL NUMBER :M.02213

Automatic Depalletter: Phase 2

**General Description and Function**

*This second stage completes the separation of the tile from the mould.

*Comprising 6 non-driven blade discs (Ø120 mm), this section operates via push force from the preceding tile/mould unit.

*Its correct function is highly dependent on the proper operation of the first (pre-demolding) stage.

Critical Points and Risk Management**Operating Principle:**

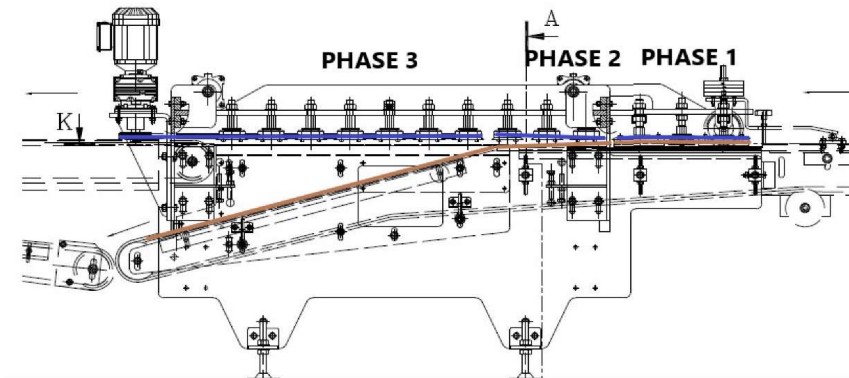
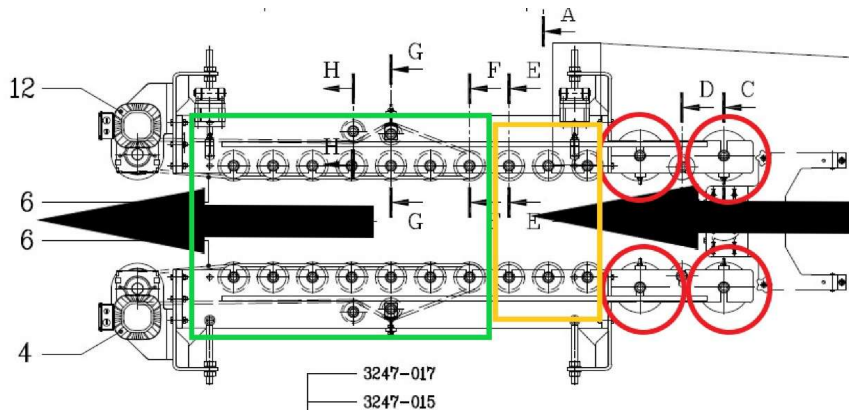
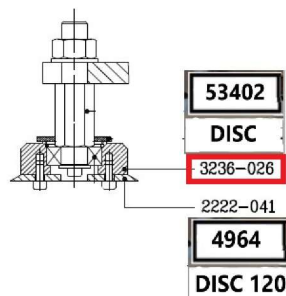
*The tile rides and is transported along the top cutting edge of the blades. Consistent blade geometry is therefore critical for smooth transit.

Key Mechanical Requirement:

*Maintaining precise parallelism between all 6 blade discs is essential to prevent tile binding, misalignment, or damage during the final separation.

Primary Risk:

*Partial adhesion or mis-separation can occur if the pre-demolding stage is ineffective or if these blades are worn, leading to production stoppages or tile damage.

**SECTION E-E****SUPERVISION LEVEL****DAILY**

53402 Disc ø120 Blade (6 units)

4964 Disc (6 units)

Visual inspection for wear and damage.

*Check the wear condition of the 6 blade discs (Ø120 mm). Focus on the sharpness and integrity of the top edge where the tile travels.

WEEKLY

19517 Bearing CUS 62205 2RSR (6 units)

Inspection for leaks, noise, and vibration.

*Visually inspect the bearings listed under daily supervision for signs of damage or misalignment.

CLEANING**WEEKLY**

*Clean all surfaces of the structure, blades and discs.

*Use a compressed air gun with low/controlled pressure and dry cloths.

AUTOMATIC DEPALLETTER**MACHINE TYPE: TM-107 SERIAL NUMBER :M.02213****Automatic Depalletter: Phase 3****General Description and Function**

*This third section is responsible for guiding the tile through 14 motorized blade discs (Ø120 mm), which are laterally driven by a Trapezoidal (V) Belt system.

*Its function is to receive the tile from the demolding stage and deposit it smoothly onto the Depalletter's output tile conveyor.

Critical Points and Risk Management**Outfeed:**

*The height of the exit polycord conveyor should be adjusted on one or both sides, based on differences in the bottom contour of the tiles for the new profile.

Controlled Transfer:

*The transfer of the tile from the demolder to the output conveyor must be carefully controlled to prevent tile tipping, misalignment, or breakage.

Key Mechanical Requirement:

*Maintaining precise parallelism between both sets of the 14 motorized discs is critical to ensure smooth, straight-line tile transfer without binding.

Drive System Integrity:

*The V-belt drive system (tension and wear) is essential for consistent speed and synchronization of the motorized blades.

SUPERVISION LEVEL**DAILY**

53402 Disc ø120 Blade (14 units)

4964 Disc (14 units)

240677 V-Type belt (2 units)

Visual inspection for wear and damage.

*Check the wear condition of the 14 motorized blade discs (Ø120 mm). Inspect the guiding edges.

*Check for oscillation or excessive play in the 14 blade disc bearings and the 4 belt tensioner bearings.

*Inspect the 2 V-belts for wear (cracks, fraying)

*Check and adjust the tension of the 2 V-belts to the manufacturer's specification.

WEEKLY

19517 Bearing CUS 62205 2RSR (14 units)

17999 Bearing CUS 6205 2RS1 (4 units)

58971 Disc (2 units)

58975 Disc (2 units)

289575 Gear-box + Motor (1 u)

289576 Gear-box + Motor (1 u)

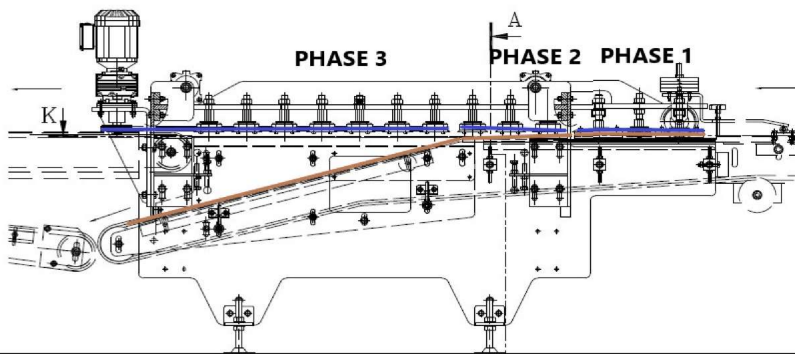
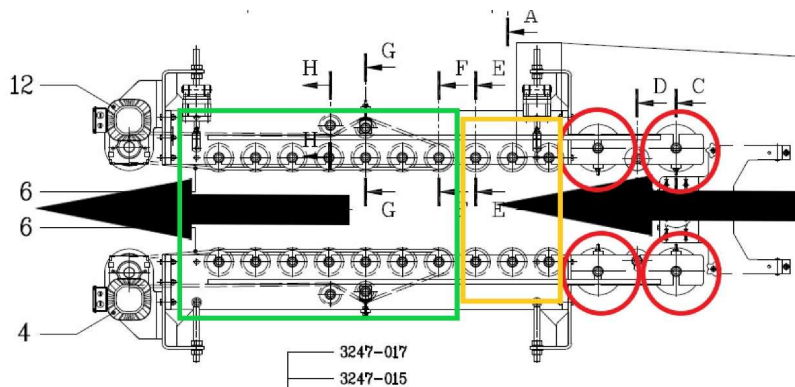
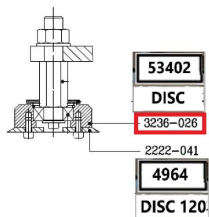
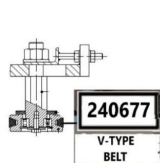
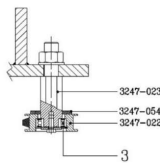
Inspection for leaks, noise, and vibration.

*Visually inspect the bearings listed under daily supervision for signs of damage or misalignment.

CLEANING**WEEKLY**

*Clean all surfaces of the structure, blades, discs, and the V-belt drive area.

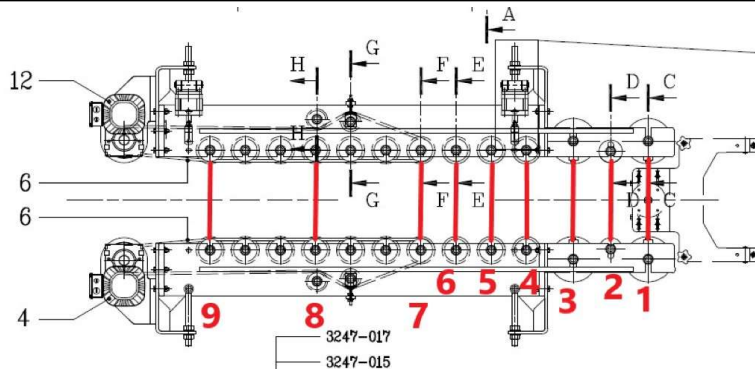
*Use a compressed air gun with low/controlled pressure and dry cloths.

**SECTION E-E****SECTION G-G****SECTION H-H**

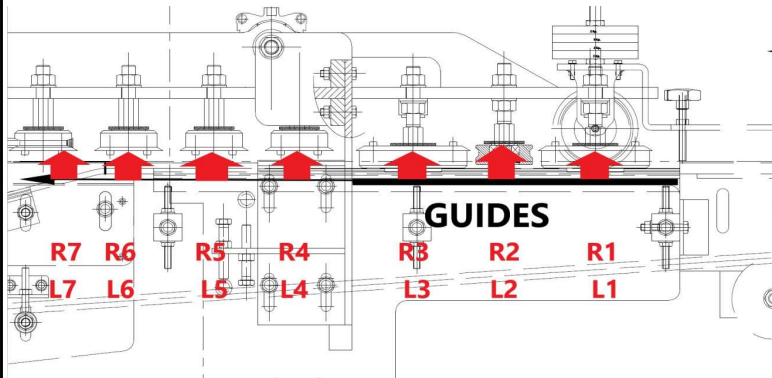
AUTOMATIC DEPALLETTER

MACHINE TYPE: TM-107 SERIAL NUMBER :M.02213

Automatic Depalletter: Positioning and Alignment



MEASURE	DATE :								
DISCS	1	2	3	4	5	6	7	8	9



*The Height positioning measurements for the reference 7 Blades.

RIGHT (R) LEFT (L)

MEASURE	DATE :						
HEIGHT	L1	L2	L3	L4	L5	L6	L7
HEIGHT	R1	R2	R3	R4	R5	R6	R7

General Description and Function**POSITIONING AND ALIGNMENT**

*To ensure uniform and trouble-free tile demolding, precise reference positioning for all discs and blades must be established and maintained

*This is critically important, as the first 12 discs on the support assembly have multi-directional adjustment capability can shift over time.

Critical Points and Risk Management

*The depalletter must maintain a single, fixed setting for each of the tile profiles in production.

*Consistency in width and height alignment is paramount to demolding issues.

Width Positioning:

*Check and document the width positioning measurements for the established reference discs.

*Compare against the master reference values. Any deviation greater than $\pm X$ mm requires correction and re-alignment.

Height Alignment Check

*Reference Blades (7 units)

*Verify and record measurements.

*Distinguish between Right (R) and Left (L) sides.

*Measurements for the established reference discs.

*Compare against the master reference values. Any deviation greater than $\pm X$ mm requires correction and re-alignment.

Height Positioning:

*Check and document the height positioning measurements for the 7 reference blades.

*Crucially, perform this check for both the RIGHT (R) and LEFT (L) sides to ensure perfect symmetry and parallelism.

*Compare against the master reference values. Any deviation greater than $\pm X$ mm requires correction and re-alignment.

WEEKLY or EVERY TWO WEEKS**Width Alignment Check**

*Reference Discs

*Verify and record positioning measurements.

Height Alignment Check

*Reference Blades (7 units)

*Verify and record measurements.

*Distinguish between Right (R) and Left (L) sides.